Thermal-Aware Power Budgeting for a Quad-Core RISC-V Processor

Anamika M (13) • Gautham D (30) • Geojin Mathew (31) • Sidharth P (62) • Sreya Manoj (66)

Department of Electronics and Communication Engineering
Saintgits College of Engineering, Kerala, India
PDEs, Transforms & Optimization Techniques, Capstone Project

***Abstract*—Thermal-aware power budgeting determines how a fixed chip power budget should be distributed to keep the hottest point as cool as possible. This study models a $10\,\mathrm{mm} \times 10\,\mathrm{mm}$ die containing four RISC-V cores and a shared central L2 cache, with a total power budget of $5\,\mathrm{W}$ and all die edges fixed at $300\,\mathrm{K}$. Fourier's law and energy conservation lead to a heat-conduction PDE; the Laplace transform connects transient thermal storage to steady-state thermal impedance; finite differences produce a sparse grid model; and a linear program minimizes the maximum nodal temperature rise. The reported optimum allocates $1.175\,\mathrm{W}$ to each core and $0.300\,\mathrm{W}$ to the cache. In the all-cores-active case, the peak falls from $325.18\,\mathrm{K}$ to $322.86\,\mathrm{K}$; in the two-idle-core comparison, it falls from $328.65\,\mathrm{K}$ to $322.86\,\mathrm{K}$, a stated reduction of $5.79\,\mathrm{K}$. The symmetric four-point validation case shows no improvement, consistent with symmetry. These are simulation results based on synthetic power and idealized steady-state assumptions, not measurements of fabricated silicon.**

***Index Terms*—thermal management, heat equation, Laplace transform, finite differences, linear programming, RISC-V.**

I. Introduction

A multicore processor converts electrical power into heat within a small silicon area. Heat conducts through the die toward a heat sink, and the maximum temperature can determine throttling behaviour, leakage and long-term reliability. Consequently, the average temperature alone is not a sufficient design target: the hottest location must be considered directly.

The project asks how to distribute a fixed $5\,\mathrm{W}$ power budget over a $10\,\mathrm{mm} \times 10\,\mathrm{mm}$ die so that its peak temperature is minimized. The die edges are held at $300\,\mathrm{K}$. Unlike a generic four-block layout, the processor embodiment places four cores in the quadrants and a shared L2 cache at the centre. The central cache is farther from the cooled boundary and can therefore have a different thermal effect from an equally powered core. The study builds a connected mathematical pipeline from physical heat flow to constrained optimization rather than treating optimization as a separate trial-and-error step.

II. Our Embodiment

The core design is based on FRISC-V, an open-source RV32I processor described as having a five-stage pipeline [6]. The die arrangement used for the model is schematic; the core and cache rectangles are simplified footprints, not a detailed physical floorplan. The original symmetric four-point layout from the assignment is retained as a separate validation case and its results are not combined with the processor-layout results.

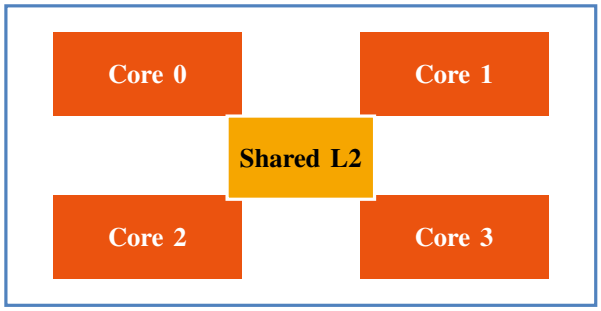


Fig. 1. Schematic die model. All four edges are held at $300\,\mathrm{K}$; the central L2 cache is a separate power-allocation block.

III. Related Work

Carslaw and Jaeger provide the classical analytical foundation for steady and transient heat conduction [1]. Their solutions establish the governing physics, although closed-form solutions are restricted to idealized geometries and do not directly determine a multi-block power allocation. Huang *et al.* introduced HotSpot, a compact thermal modelling methodology that represents floorplan blocks and package effects through a thermal network [2]. Such models are efficient, but a block-level network can compress spatial detail. Murali *et al.* studied temperature-aware frequency assignment for multiprocessor systems-on-chip [3], while Boyd and Vandenberghe describe the convex-optimization framework used to formulate global minimax problems [4].

The approach here retains a two-dimensional spatial field and places its discretized heat operator directly inside a linear program. This is a deliberately simplified educational model, not a replacement for validated compact or finite-element thermal tools. Its contribution is the explicit connection between the PDE-derived matrix and the optimizer.

IV. Methodology

The methodology links five stages: heat-flux physics, a boundary-value PDE, a Laplace-domain interpretation, finite-difference discretization, and linear programming.

A. Field model and governing PDE

Fourier's law states that heat flux moves down the temperature gradient:

$$\mathbf{J} = -k\nabla T, \tag{1}$$

where $\mathbf{J}$ is heat flux in $\mathrm{W/m^2}$, $k = 150\,\mathrm{W/(m\,K)}$ is the assumed constant thermal conductivity, T is temperature in kelvin, and ∇T is the spatial temperature gradient. The minus sign indicates that heat flows from hot regions toward cold regions. At steady state, conservation of energy gives

$$\nabla \cdot \mathbf{J} = Q, \tag{2}$$

where Q is volumetric heat generation in $\mathrm{W/m^3}$. Substituting (1) into (2), and taking k as constant, gives the Poisson equation

$$\nabla^2 T = -\frac{Q}{k}. \tag{3}$$

Here $\nabla\cdot$ is divergence and ∇^2 is the Laplacian, which describes spatial curvature of the temperature field.

Let $\theta = T - T_{\mathrm{amb}}$ be temperature rise above ambient, where $T_{\mathrm{amb}} = 300\,\mathrm{K}$. Prescribing the edge temperature gives the Dirichlet boundary condition $\theta = 0$ on $\partial\Omega$, the boundary of the square die. The model assumes a thin plate with temperature uniform through its thickness, constant conductivity, adiabatic top and bottom faces, and ideal isothermal edges. These assumptions define a steady, elliptic boundary-value problem.

B. Laplace transform and thermal impedance

To connect the steady model with transient heating, thermal storage is included:

$$\rho c_p \frac{\partial \theta}{\partial t} = k\nabla^2\theta + Q, \tag{4}$$

where ρ is density, c_p is specific heat capacity, and t denotes time. The product ρc_p is volumetric heat capacity. Applying the Laplace transform in time, $\Theta(x,y,s) = \int_0^\infty e^{-st}\theta(x,y,t)\,dt$, converts the time derivative to $s\Theta - \theta(x,y,0)$. For an initially ambient die, $\theta(x,y,0) = 0$, so the transformed equation is

$$\rho c_p s\Theta = k\nabla^2\Theta + \widehat{Q}. \tag{5}$$

The complex variable s has units of inverse seconds, and the hat denotes a transformed source. Thermal impedance from power input P_j in block j to temperature rise at observation point o is $Z_{oj}(s) = \Theta_o(s)/P_j(s)$, measured in K/W. For a stable system, the final-value theorem relates the steady response to the low-frequency limit of the thermal transfer function. In that limit, the storage contribution no longer affects the steady-state input–output relation, which is governed by the steady conduction equation. Thus, the optimizer uses the zero-frequency thermal response; it does not optimize transient workloads.

C. Finite-difference discretization

The square is represented by a 40×40 grid. A second derivative is approximated by a centred difference, for example $\partial^2\theta/\partial x^2 \approx [\theta_{i+1,j} - 2\theta_{i,j} + \theta_{i-1,j}]/h^2$, with grid spacing h. Combining the x- and y-direction approximations gives the five-point stencil

$$\frac{4\theta_{i,j} - \theta_{i+1,j} - \theta_{i-1,j} - \theta_{i,j+1} - \theta_{i,j-1}}{h^2} = \frac{Q_{i,j}}{k}. \tag{6}$$

Boundary nodes are fixed to zero temperature rise. The matrix assembled from the stencil is sparse because each interior node couples only to its four neighbours. For a block b with power q_b distributed uniformly over n_b grid cells and die thickness t_z, the source density in each covered cell is $Q_{i,j} = q_b/(n_b h^2 t_z)$. Collecting nodal temperatures into vector $\boldsymbol{\theta}$ and the five block powers into $\mathbf{q}$ gives

$$A\boldsymbol{\theta} = B\mathbf{q}, \qquad \boldsymbol{\theta} = M\mathbf{q}, \quad M = A^{-1}B. \tag{7}$$

Here A is the discretized conduction operator, B maps block powers to grid sources, and M is the block-power-to-temperature-rise map. The implementation described in the project uses a 1600×1600 sparse matrix; an interior-only formulation would instead eliminate the boundary nodes. The exact thickness and footprint dimensions should be confirmed against the repository implementation because they are not specified in the presentation.

D. Linear-programming formulation

The peak rise is $\max_i \theta_i$. Introduce an auxiliary variable u that bounds every nodal rise. The optimization is

$$\begin{aligned} \min_{\mathbf{q},\boldsymbol{\theta},u} \quad & u \\ \text{subject to} \quad & A\boldsymbol{\theta} = B\mathbf{q}, \\ & \theta_i \le u \quad \text{for every grid node } i, \\ & \sum_{b=1}^{5} q_b = 5, \quad q_b \ge 0, \quad q_{\mathrm{L2}} \ge 0.3. \end{aligned} \tag{8}$$

The decision variables q_b are the powers assigned to the four cores and cache in watts; θ_i and u are temperature rises in kelvin. The equality enforces the full 5 W budget, while non-negativity prevents negative power and the cache floor enforces its minimum allocation. Because the thermal matrix is linear and every constraint is linear, (8) is a linear program. Minimizing u minimizes the largest nodal rise and hence the peak temperature. The reported implementation solves the LP with SciPy's HiGHS solver.

V. Experimental Setup

The work is a software simulation, not a physical experiment. The described pipeline uses Python 3, NumPy, SciPy sparse-matrix routines [5], `scipy.optimize.linprog` with HiGHS, and Matplotlib. A Tkinter dashboard is described as reusing the batch solver functions. The project presentation reports that one command, `python run.py`, executes the pipeline, that 16 of 16 unit tests pass, and that a modal

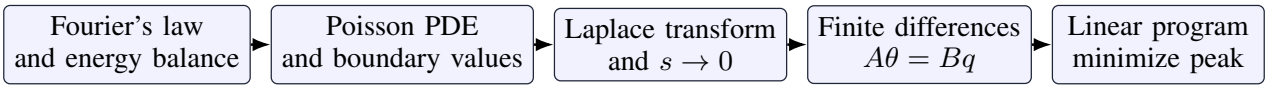


Fig. 2. Connected mathematical pipeline: each stage provides the model or quantity used by the next.

TABLE I
PRINCIPAL MODEL PARAMETERS AND THEIR STATUS.

Quantity	Value	Status
Die size	10 mm × 10 mm	Assignment
Conductivity	150 W/(m K), constant	Presentation
Edge temperature	300 K on all edges	Assignment
Total power	Sum of block powers = 5 W	Assignment
L2 minimum	At least 0.3 W	Model constraint
Grid	40 × 40 (1600 nodes)	Presentation
Heat sources	Uniform within block footprints	Assumption

and sparse-solver comparison differs by 5.91×10^{-16}. These are reported implementation checks; they do not constitute validation against measured silicon or an independent thermal simulator.

Three cases are compared: all four cores active; cores 2 and 3 idle in the uniform baseline; and the original symmetric four-point layout used for validation. Each reported peak is compared against the stated uniform split. The precise definition of the uniform allocation and the geometry parameters should be cross-checked with the repository before final submission.

VI. Results and Discussion

Table II gives the peak temperatures reported in the presentation. In Scenario A, the LP allocation reduces the peak from 325.18 K to 322.86 K, with the presentation stating a 2.31 K reduction. However, the displayed temperatures subtract to $325.18 - 322.86 = 2.32$ K. This report uses 2.32 K when calculating from the displayed temperatures and flags the presentation value for confirmation against unrounded output. In Scenario B, the uniform baseline reaches 328.65 K, while the LP reaches 322.86 K, a 5.79 K reduction. The symmetric validation case remains at 343.45 K for both allocations, so no reduction is observed.

TABLE II
PEAK DIE TEMPERATURE AT A 5 W BUDGET, USING VALUES REPORTED IN THE PRESENTATION.

Case	Uniform (K)	LP (K)	Stated Drop (K)
A: all cores active	325.18	322.86	2.32
B: two cores idle	328.65	322.86	5.79
Symmetric validation layout	343.45	343.45	0.00

The presentation reports the same optimized plan for Scenarios A and B: 1.175 W per core and 0.300 W for the L2 cache. This totals $4(1.175) + 0.300 = 5.000$ W. The cache stays at its minimum allocation, while the equivalent cores share the remainder equally. The result is consistent with reducing power at the centrally located, comparatively

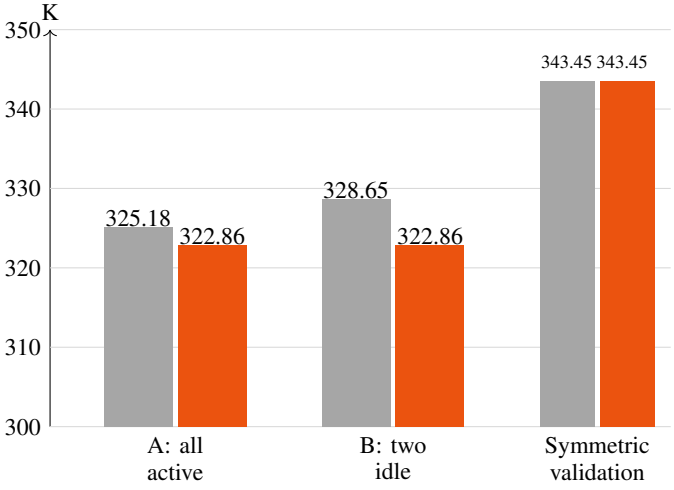


Fig. 3. Reported peak temperatures. The vertical scale begins at 300 K, the imposed edge temperature.

poorly cooled cache. Scenario B must be interpreted carefully: the reported optimum still allocates power to all four cores, including the two described as idle in the uniform baseline. Therefore, the 5.79 K difference compares the stated baseline with a thermally balanced allocation; it is not proof that a real controller could deliver that gain while keeping those cores idle.

For the symmetric four-point case, uniform allocation is already optimal under the stated symmetric model. No gain is therefore expected, and the zero reduction is a useful sanity check. The presentation also reports 4.03 K/W versus 5.73 K/W for peak rise per watt, but 4.03 K/W cannot be reproduced from the listed LP peak $(322.86 - 300)/5 = 4.57$ K/W. That value should be verified against the code rather than silently reconciled.

VII. Limitations and Conclusion

The model uses synthetic block powers rather than measurements from FRISC-V silicon. It assumes constant conductivity, a two-dimensional thin plate, ideal 300 K boundaries, simplified rectangular footprints, and no package or heat-spreader layers. The die thickness and exact footprint dimensions are not stated in the presentation, although they can materially affect absolute temperatures. The 0.3 W cache floor also influences the optimum. Because the optimization is steady-state, it does not constrain short power bursts or time-varying workloads. Internal unit tests and solver agreement are useful checks, but independent validation is still needed.

The mathematical pipeline shows how vector calculus, PDEs, Laplace transforms, numerical discretization and linear

TABLE III
PIPELINE CONTRIBUTION RECORD

Stage	Contribution to record	Assigned Member
Field model and PDE	Fourier law, energy balance, boundary conditions	Sidharth P
Laplace transform	Transient equation and steady-state interpretation	Sreya Manoj
Discretization	Grid, five-point stencil, sparse matrix	Geojin Mathew
Optimization	LP variables, constraints, solver	Anamika M
Results and report	Figures, interpretation, final checks	Gautham D

programming can be combined into one engineering workflow. Within the reported model, the LP reduces the peak by $2.32\,\mathrm{K}$ based on the displayed temperatures in the all-active case and $5.79\,\mathrm{K}$ in the two-idle-core baseline comparison, while the symmetric validation case shows no improvement. These results support the method's internal logic, not a claim of measured real-chip performance. Future work should use measured block power, code-confirmed floorplan dimensions, a three-dimensional package and heat-spreader model, temperature-dependent conductivity, transient workloads, and validation against an established thermal simulator.

REFERENCES

[1] H. S. Carslaw and J. C. Jaeger, *Conduction of Heat in Solids*, 2nd ed. Oxford, U.K.: Oxford University Press, 1959.

[2] W. Huang *et al.*, "HotSpot: A compact thermal modeling methodology for early-stage VLSI design," *IEEE Trans. Very Large Scale Integr. (VLSI) Syst.*, vol. 14, no. 5, pp. 501–513, May 2006.

[3] S. Murali, A. Mutapcic, D. Atienza, R. Gupta, S. Boyd, and G. De Micheli, "Temperature-aware processor frequency assignment for MPSoCs using convex optimization," in Proc. 5th IEEE/ACM/IFIP Int. Conf. Hardware/Software Codesign and System Synthesis (CODES+ISSS), Salzburg, Austria, 2007, pp. 111–116, doi: 10.1145/1289816.1289845.

[4] S. Boyd and L. Vandenberghe, *Convex Optimization*. Cambridge, U.K.: Cambridge University Press, 2004.

[5] P. Virtanen *et al.*, "SciPy 1.0: Fundamental algorithms for scientific computing in Python," *Nature Methods*, vol. 17, pp. 261–272, 2020.

[6] FRISC-V, "FRISC-V: open-source RV32I processor," in repository

Repository: Capstone Project Group 7 GitHub repository